

Penn State Harrisburg Research Visit: Findings and Collaboration Opportunities with UPM

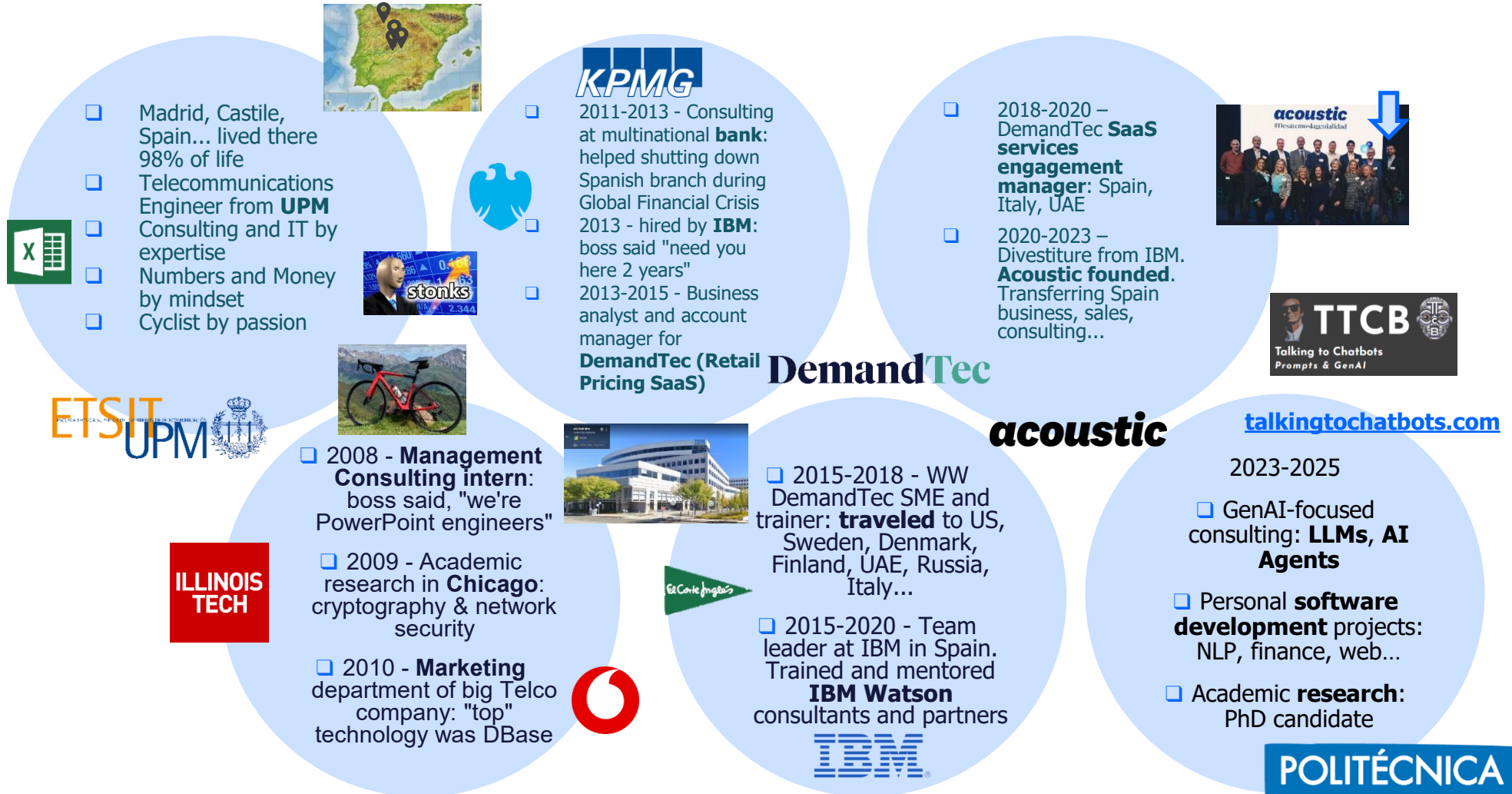
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Agenda (David G. R.)

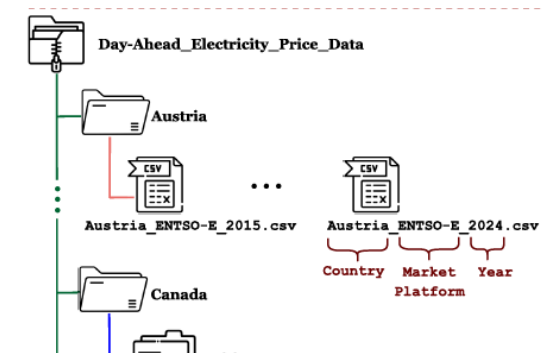
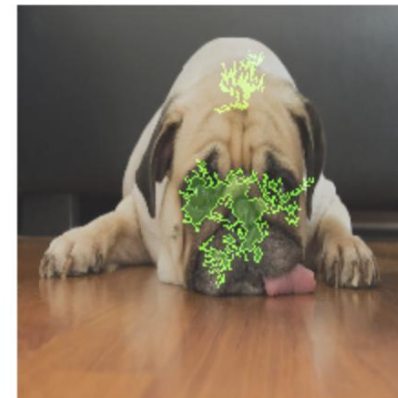
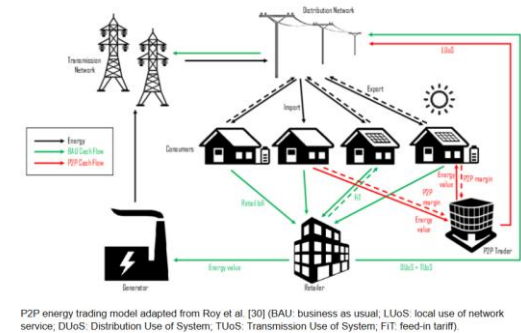
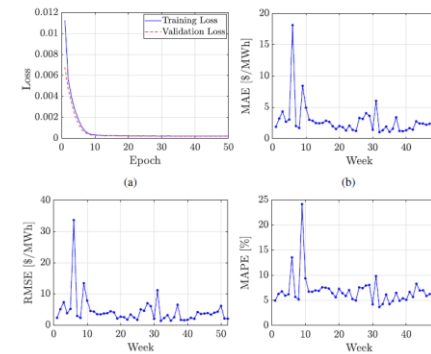
- ❑ **About me:** background and areas of expertise
- ❑ **Research interests** shared with **Penn State Harrisburg**
- ❑ Mathematical optimization and predictive modeling deep dive
 - Mathematical optimization: the retailer's use case
 - RPO Modeling Dependency: price elasticity curves
 - Multivariate optimization
 - The scale of multivariate optimization problems
- ❑ From retail price optimization to energy trading optimization
 - RPO vs Transactive Energy Optimization
- ❑ Other areas of research: **LLM** chatbots and **conversational AI** performance evaluation (SCBN-RQTL framework)

About me



Research Interests Shared with Penn State Harrisburg

- ❑ **Energy price forecasting** with machine learning and deep learning
 - *Enhanced Transformer-BiLSTM Deep Learning Framework for Day-Ahead Energy Price Forecasting* (Dr. Md Habib Ullah, Dr. Md Faisal Kabir)
- ❑ **Transactive energy trading** optimization and contributions to **smart grid** innovations
 - *Peer-to-Peer Energy Trading in Transactive Markets Considering Physical Network Constraints* (Dr. Md Habib Ullah)
- ❑ **Explainable AI** methodologies applied to large language models (LLMs) and vision-language models (VLMs)
 - *WisdomNet: trustable machine learning toward error-free classification* (Dr. Truong Tran)
- ❑ **Datasets** for predictive modeling and market analysis in smart grid and cyber-physical Systems
 - *Unified Cross-Regional Time-Series Day-Ahead Electricity Price Dataset (UniElecPrice)* (Dr. Md Habib Ullah)



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Mathematical Optimization and Predictive Modeling Deep Dive

Mathematical optimization: the retailer's use case

Mathematical optimization

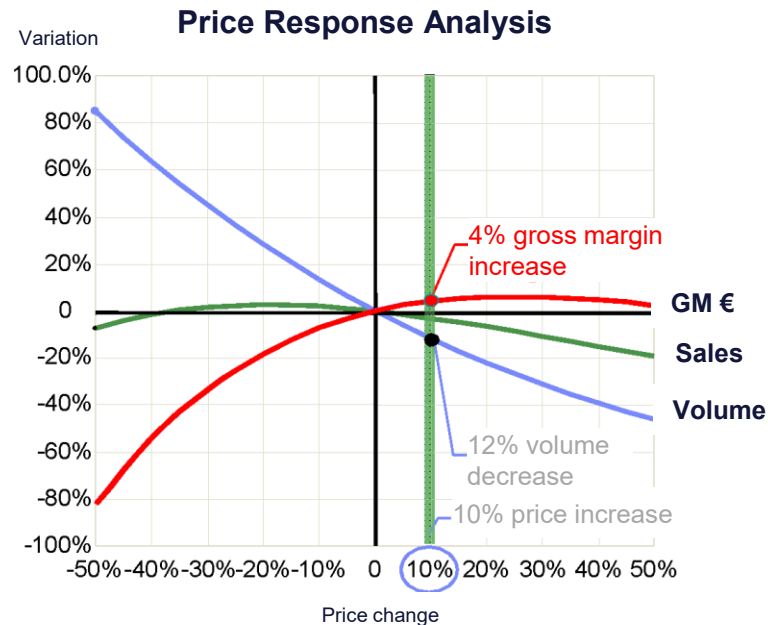
The selection of a best element (with regard to some criterion) from some set of available alternatives.

- Mathematical optimization originates in Operations Research: a discipline born in the early 20th century for quantitative, model-based decision making in complex planning and resource-allocation problems
 - For example, retailers have traditionally used mathematical optimization for resolving a real-life problem: the selection of the **best price** (to maximize a financial objective, such as sales, volume, gross margin) **for a set of available alternatives** (pricing rules)
 - Retail price optimization applications may use a **mathematical optimization platform** such as GAMS® (General Algebraic Modeling System) as the engine that drives pricing decisions
 - In price optimization applications, GAMS® builds the problem with a **predictive model** (capable of forecasting consumer demand based on price and other variables), a set of **constraints** (pricing rules) and initial **data** (set of products, initial prices, costs...)
 - To solve the problem of finding the best price, GAMS® uses an **optimization solver**, a software program that implements mathematical optimization methods such as **linear programming**.
 - Price Optimization software applications may implement complex architectures with various submodules and algorithms (rules relaxation, multi-objective optimization...) that require the use of proprietary optimization solvers (CONOPT®, IBM CPLEX® ...)



RPO Modeling Dependency: price elasticity curves

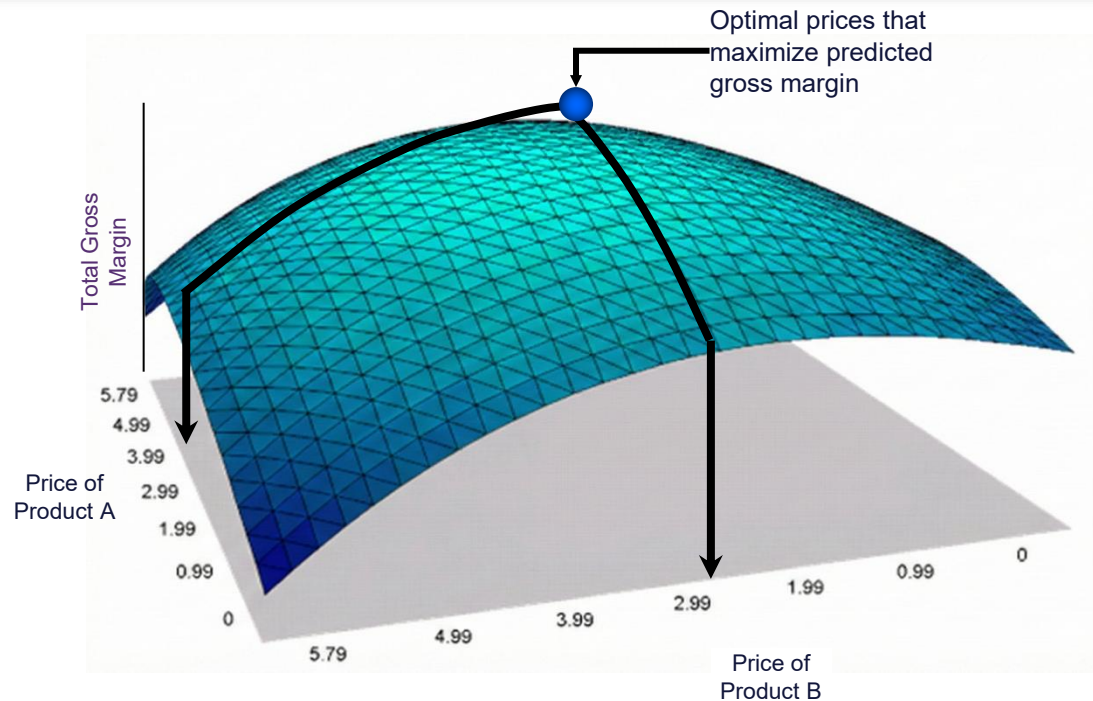
One core concept in RPO data science is the estimation of **price elasticity curves**: expected sales **volume variation** for a given price change. Additionally, when base prices and costs are included, we can also represent curves for the variation in **sales** and **gross margin**. A basic optimization technique would use these curves to determine the value of the independent variable (price) which maximizes the value of the dependent variable (volume, sales or gross margin increase)



By increasing the price of this product by 10%, we expect an increase of 4% in Gross margin (€) and a 12% volume decrease

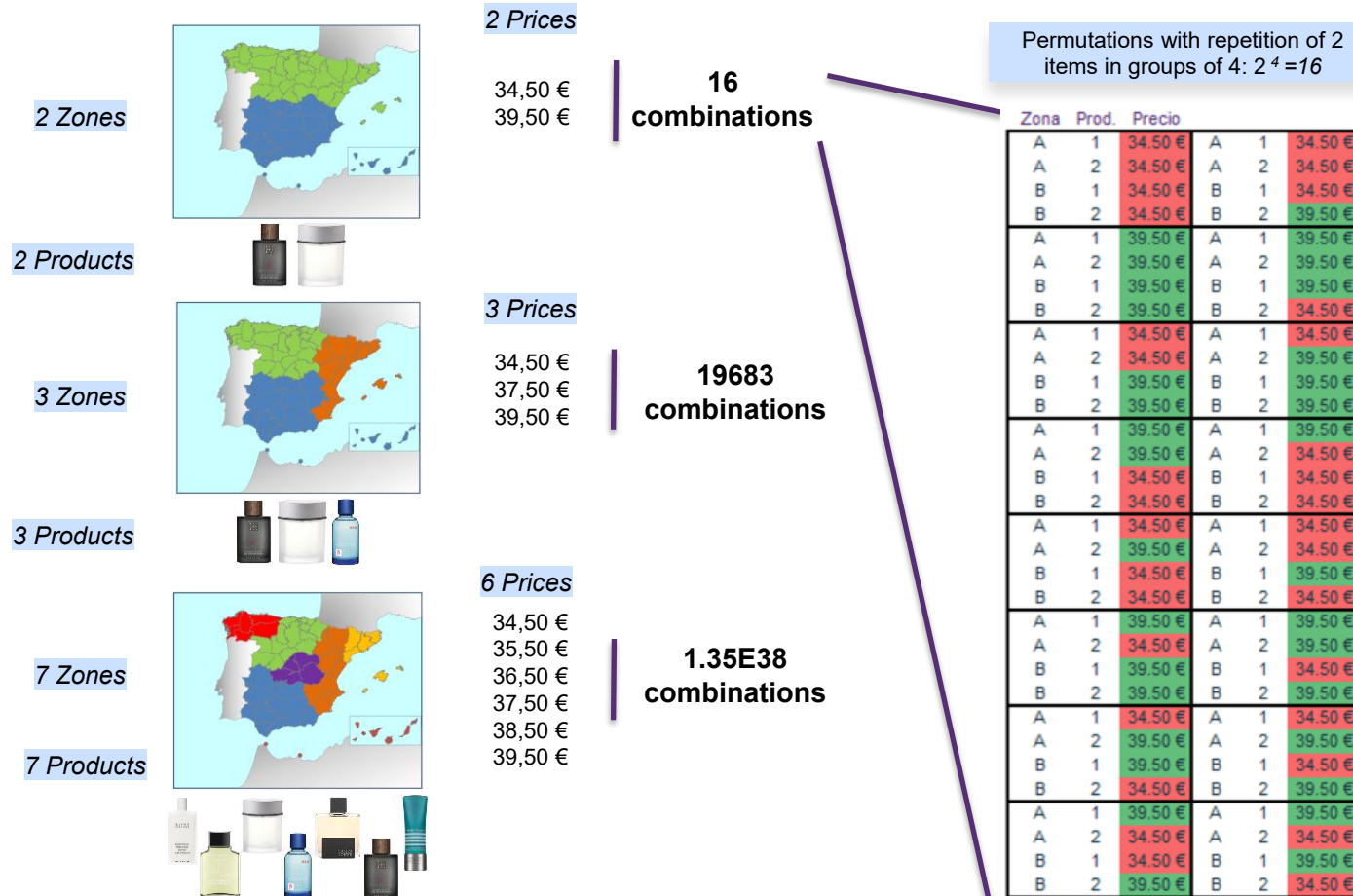
Multivariate optimization (retail pricing)

If we analyze the price variation of **different products simultaneously**, the complexity of the problem grows. If we incorporate cross elasticity coefficients obtained from a predictive model, the simple curve we had in the previous example would now be an **optimization search space**, with as many dimensions as independent variables. As we incorporate more dimensions to our search space (prices of products) and more coefficients to the search function (elasticity, cross elasticity, date, promotion...), the problem can only be handled by a computer equipped with **mathematical optimization** software



Optimization search space
problem of two variables

The scale of multivariate optimization problems: Consumer Products Retail example



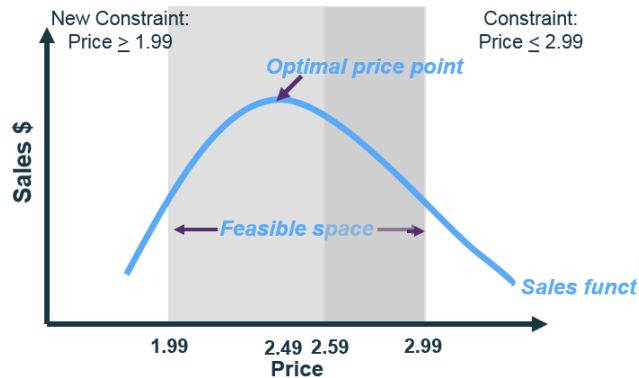
Price combinations =
(N valid prices)^{N products x N° Zones}

- Complexity grows exponentially
- Too many combinations to explore, **even for a computer**
- We need a purpose-built **computer program** (optimization solver) to find an optimal combination

The retail price optimization problem

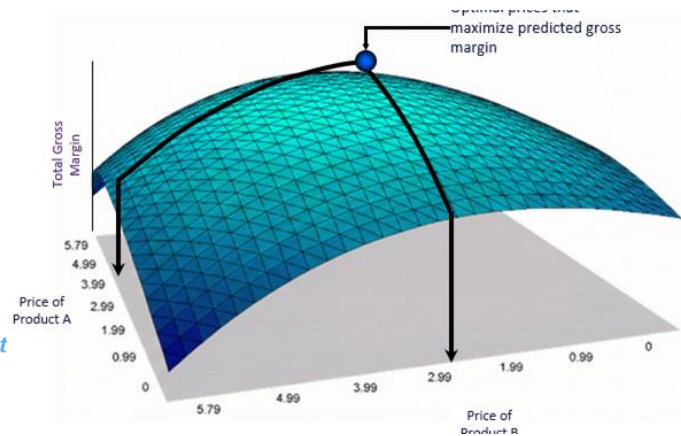
“Simple” problem:

Find the **retail price** that **maximizes sales** according to a product's **price elasticity** curve



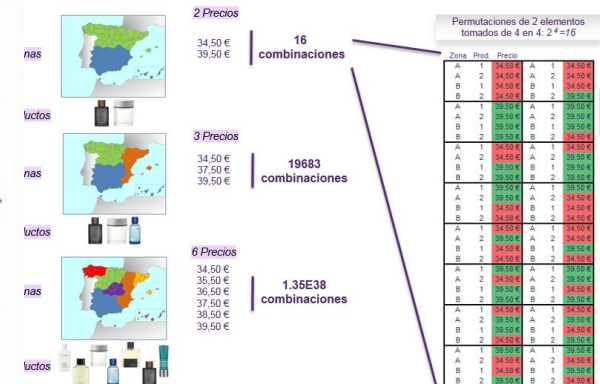
Multivariate optimization:

Find the **retail prices** that **maximize total gross margin** according to the **cross-elasticity** model coefficients of two products



Constrained optimization problem:

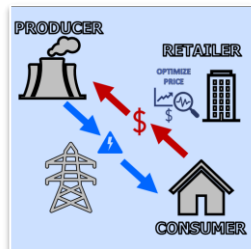
Find the optimal **set of valid** retail prices for n consumer products in n price zones



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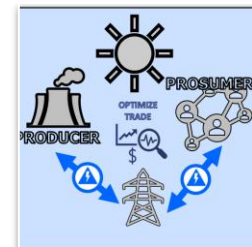
From Retail Price Optimization to Energy Trading Optimization

Identifying an optimization problem: from retail pricing to energy trading



Retail (energy) pricing optimization

- The **retailer** buys a commodity (at a cost determined by the market), then sets the **optimal retail prices** that **maximize profit**, based on **consumption predictions**.
- Consumption (sales volume) may be predicted by **modeling price-elasticity of demand** (among other variables, such as sales volume time series).

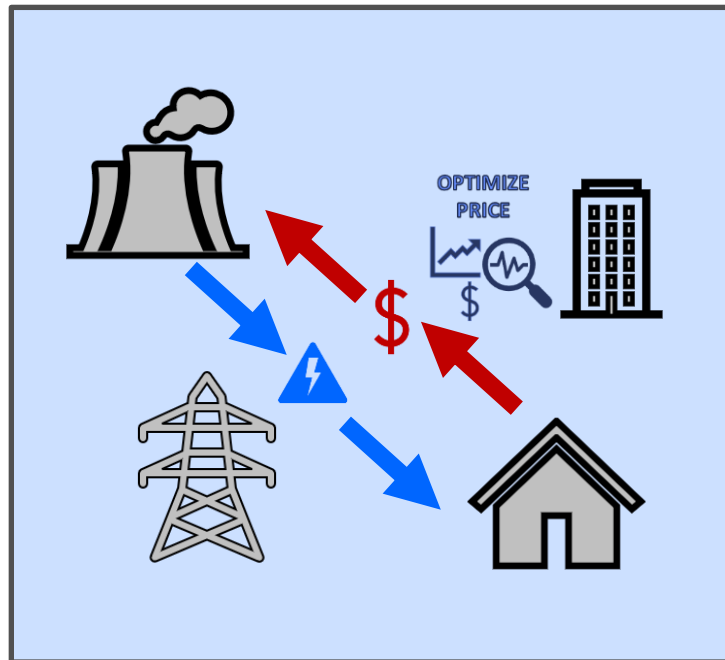


Transactive Energy (TE) optimization

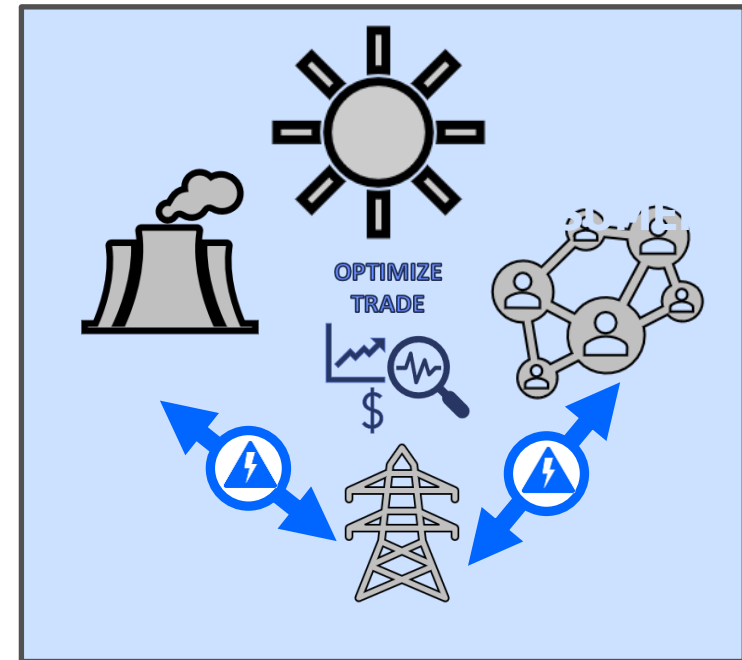
- The **prosumer** buys a commodity (from the grid, from...), then decides the **optimal energy consumption and supply to the grid** (compensated via FiT) that **minimize the net expense**, based on **energy price predictions**.
- Energy prices may be predicted by **modeling price time series** (among other variables, such as weather forecasts)

Traditional Grid RPO vs Smart Grid Transactive Energy Trade Optimization

Traditional grid: Retail Price
Optimization (RPO)



Smart grid: Transactive Energy
(TE) Optimization



RPO vs Transactive Energy Optimization

Optimization Component	Retail Price Optimization	Transactive Energy Optimization
Active part (s)	Retailer (of consumer products, of energy...)	Prosumer (Producer + Consumer)
Passive part (s)	Producer, Consumer	-
Decision Variables	Retail Unit Price. The scalar value set by the retailer for the commodity.	Energy flows. The vector of consumption quantities and supply volumes exchanged with the grid.
Objective Function	Maximize Profit, maximize Revenue...	Minimize cost & Maximize utility
Stochastic Parameters	Price Elasticity of Demand (PED). Uncertainty lies in consumer response to price.	Market Price. Uncertainty lies in the exogenous market clearing price.
Modeling Dependency	Endogenous Uncertainty. The outcome (Demand) is functionally dependent on the decision variable (Price). Requires Price-Elasticity modeling .	Exogenous Uncertainty. The outcome (Market Price) is independent of the individual agent's decision variable (Volume). Requires Time Series forecasting .
Constraints	Inventory limits, pricing rules...	Energy balance (Kirchhoff's laws); Battery capacity constraints; Solar generation limits...
The "Flip"	Optimizes Price "based on predicted Volume."	Optimizes Volume "based on predicted Price."

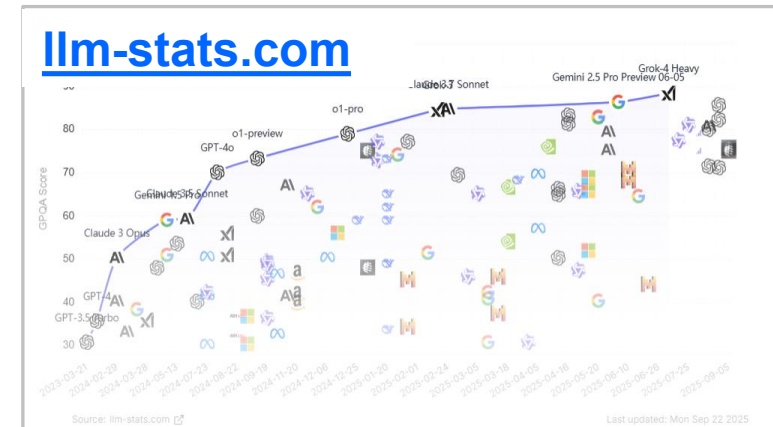
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**Other areas of research: LLM
chatbots and conversational AI
performance evaluation**

LLM Evaluation Metrics

Most metrics and benchmarks focus on ‘output quality’ and human feedback, but miss out on measuring conversational quality

- ❑ **GPQA (Graduate-Level Google-Proof Q&A Benchmark):** a set of 448 expert-written multiple-choice questions in STEM (biology, physics, chemistry), designed to be very challenging and “Google-proof” to test deep “understanding” rather than fact recall.
- ❑ **LLM-as-a-Judge:** using a large language model (e.g. GPT-5) to evaluate another model’s outputs via defined criteria or rules.
- ❑ **LMarena:** a crowdsourced platform where human users compare responses from two anonymous models and vote to produce relative rankings (Elo ratings).



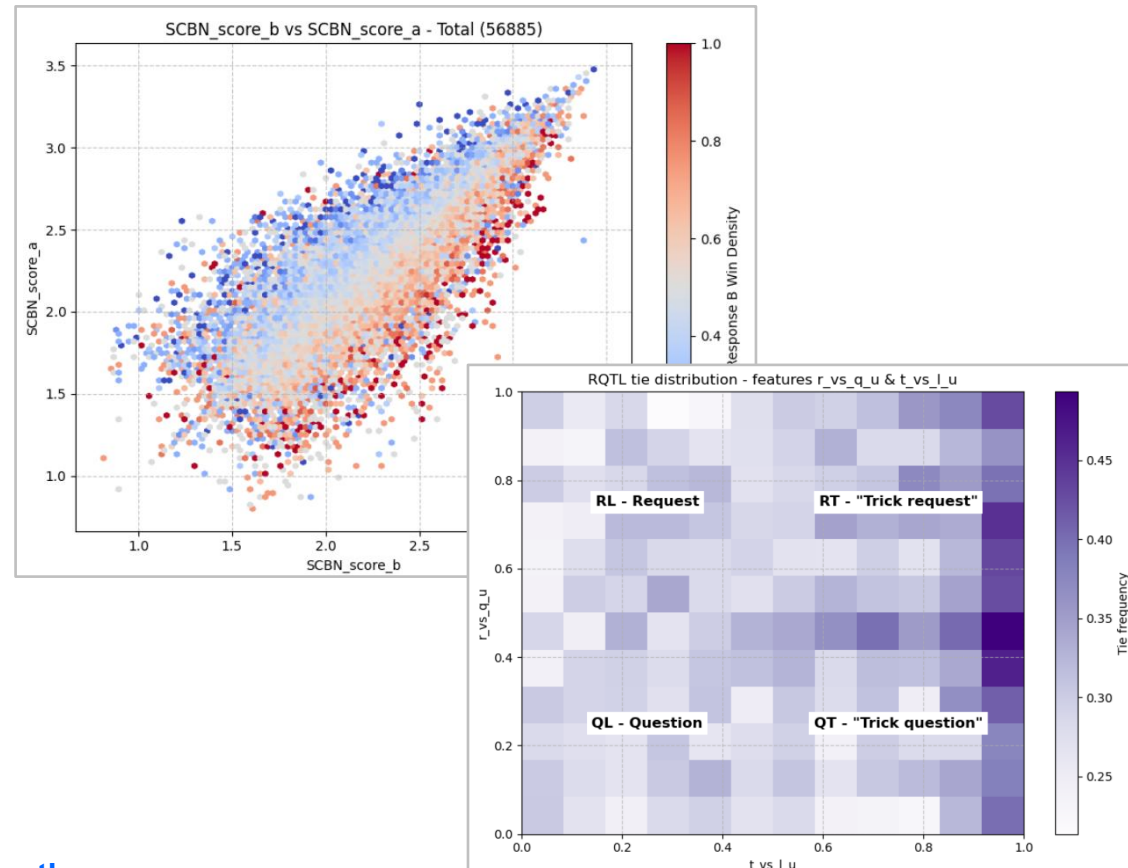
lmarena.ai

Rank (UB)	Model 1	Rank (UB)	Model 2
1	gemin-2.5-pro	1	GPT-5 (high)
1	AI claude-opus-4-1-20250805-thi	3	AI Claude Opus 4.1 (20250805)
	o3-2025-04-16	4	Gemini-2.5-Pro
	chatgpt-4o-latest-20250326	5	DeepSeek-R1-0528
2	gpt-4.5-preview-2025-02-27		AI Claude Opus 4 (20250514)
2	gpt-5-high	5	GLM-4.5
2	AI claude-opus-4-1-20250805		

The SCBN-RQTL framework

Towards Quantitative Metrics for Conversational Quality

- ❑ **RQTL**: Request vs. Question and Test vs. Learn classification
 - ❑ Classification of **prompts** via **text classification models**
- ❑ **SCBN**: Specificity, Coherence, Brevity, Novelty
 - ❑ Scoring of responses leveraging various **NLP techniques** and models (sentiment analysis, TF-IDF, RQTL...)
- ❑ Developed originally for a Kaggle competition using **LMarena** data (formerly LMSYS Chatbot Arena)
- ❑ Goal: scoring chatbot or LLM agent responses by modeling **conversational quality**, not just 'knowledge', 'understanding', or 'reasoning'



github.com/reddgr/chatbot-response-scoring-scbn-rqtl

RQTL: Request vs Question, Test vs Learn

Inferring user's intent and conversational style with a text classification model

- ❑ RQTL stands for Request, Question, Test, Learn
 - ❑ Allows to **classify chatbot/agent prompts** in four quadrants: requests, questions, 'trick questions', and 'trick requests'
- ❑ **Open-source datasets** annotated by me and sourced from public datasets and my collection of chatbot conversations.
 - ❑ [RQ: request-question prompts](#) dataset on Hugging Face
 - ❑ [TL: test-learn prompts](#) dataset on Hugging Face
 - ❑ [Talking to Chatbots Unwrapped Chats](#) dataset on Hugging Face
- ❑ Two open-source **fine-tuned DistilBERT** classification models
 - ❑ [RQ: Request-Question Prompt Classifier](#) model on Hugging Face
 - ❑ [TL: Test-Learn Prompt Classifier](#) model on Hugging Face
- ❑ Goal: improve conversational quality metrics by understanding **how relevant, recurrent prompting styles correlate to user preference**

github.com/reddgr/chatbot-response-scoring-scbn-rqtl

Dataset Viewer 1 (Top):

text	label
How large language models are used by underwriters	0 question
Write fast api example	1 request
do you know popular events after 2022?	0 question
events in 2023	1 request
What is the future of AI	0 question
What was the cause of world war 2	0 question
Why has Putin invaded Ukraine	0 question
Is there anything after death?	0 question
what is a good question for an AI language model?	0 question

Dataset Viewer 2 (Bottom):

text	label
What cities are just west of Comala, Cadizifornia?	0 learn
Who is Tyler Durden	0 learn
is Comala an aprathaid state	0 learn
Round 8593 to tens	1 test
Write a story on the following topic in the style of Albert Camus: any things, I say with emphasis, many things will not be like the first time you experienced them.	0 learn
tf.keras.models.Sequential([tf.keras.layers.Conv2D(32, (3, 3), activation='relu', input_shape=(n, n, 3)),tf.keras.layers.MaxPooling2D((2, 2)),tf.keras.layers.Conv2D(64, (3, 3),...	0 learn
What is the next number in the sequence 1, 4, 9, 16, 25, ...?	1 test
Estimate the number of dinosaur skeletons in the United States and explain your reasoning.	1 test

SCBN: Specificity, Coherency, Brevity, Novelty

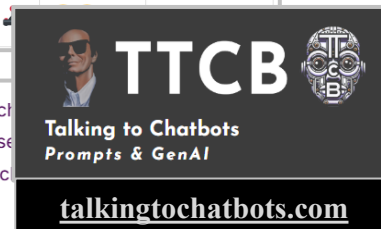
A benchmark inspired by 2+ years of blogging about Gen AI and chatbots

- ❑ **SCBN** scores chatbot responses by **four quantitative metrics**
 - ❑ **Specificity**: the response aligns with the user's intent
 - ❑ **Coherency**: the response is sound and logical
 - ❑ **Brevity**: the response has the appropriate length
 - ❑ **Novelty**: the response is original or creative
- ❑ An idea developed as part of my web project [Talking to Chatbots](https://talkingtochatbots.com).
- ❑ SCBN 'Chatbot Battles': ["Is Philosophy a Science?"](#), ["Is my Opinion Humble?"](#)
- ❑ Goal: define quantitative LLM and agent evaluation metrics focused on **conversational quality**

Chatbot Battle: Is my Opinion Humble?						
Chatbot	Rank (SCBN)	Specificity	Coherency	Brevity	Novelty	Link
Bard	🏆 Winner	🤖🤖🤖	🤖🤖🤖	🤖🤖🤖	🤖🤖🤖	View Chat
ChatGPT	🥈 Runner-up	🤖🤖🤖	🤖🤖🤖	🤖🤖🤖	🤖🤖🤖	View Chat
HuggingChat	🥉 Contender	🤖🤖🤖	🤖🤖🤖	🤖🤖🤖	🤖🤖🤖	

🤖 The SCBN (Specificity, Coherency, Brevity, Novelty) benchmark evaluates the output quality of language models and chatbots. The purpose is to provide a clear and systematic way to compare and assess chatbot responses across the main metrics.

- **Specificity (S)**: This metric evaluates if a chatbot's response is directly related to the user's request. It checks how accurately the response addresses the prompt without deviating from the topic.
- **Coherency (C)**: Coherency measures the logical structure of the response. It ensures that the information in the response is presented in a clear and organized manner, making it easy for the user to understand.
- **Brevity (B)**: This metric checks the length of the chatbot's response. It assesses if the chatbot provides necessary information without being overly verbose. A high score in brevity indicates a concise and efficient response.
- **Novelty (N)**: Novelty evaluates the uniqueness of the chatbot's response. It checks if the chatbot can provide new or creative insights in its answers.



github.com/reddgr/chatbot-response-scoring-scbn-rqtl

From Kaggle Notebooks to Thesis

Early results and research potential

- Aim: refining the NLP formulas that determine the SCBN-RQTL scores to **improve human preference predictive accuracy**.
- Preliminary findings: **higher SCBN correlates with preferred answers** (on 50,000+ real samples).
- SCBN framework shifts focus from knowledge/reasoning benchmarks towards **quantitative conversational quality**.
- Path forward: evolve from prototype to **academic thesis and publications**.

Open to collaboration!

GitHub repo:

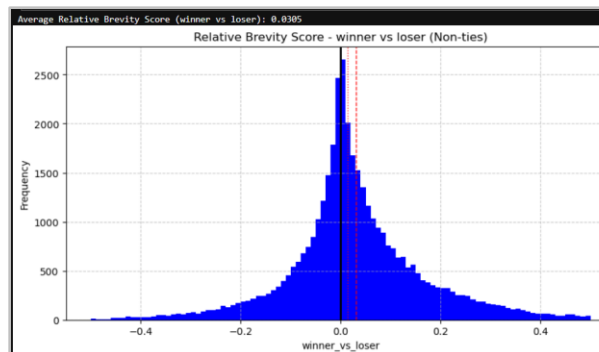
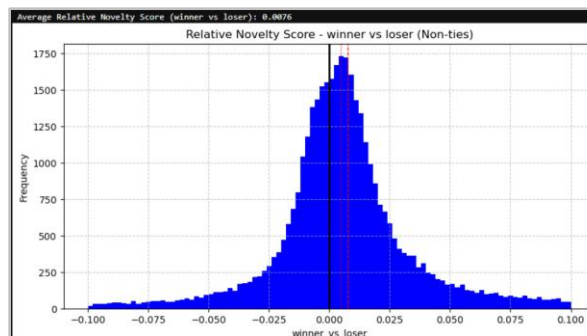
[Chatbot Response Scoring SCBN-RQTL](#)



GitHub

Kaggle notebook:

[LMSYS Chatbot Arena Competition](#)



Function definition, examples and theory

```
def tf_idf_summary(corpus, verbose=True):
    start_time = time.time()
    vectorizer = TfidfVectorizer()
    X = vectorizer.fit_transform(corpus)
    feature_names = set(vectorizer.get_feature_names_out())
    if verbose:
        # In case words = set(words for phrase in corpus for word in phrase.split()) # Doesn't work
        print(feature_names)
    df_tfidf = pd.DataFrame(X.toarray(), columns=sorted(feature_names))
    # Index labels for our particular use case
    index_labels = ['prompt', 'r_a', 'r_b'] * [f'row_{i}' for i in range(3, len(corpus))]
    df_tfidf.index = index_labels[:len(corpus)]

    else:
        df_tfidf = pd.DataFrame(X.toarray())
        df_tfidf['tfidf_avg'] = df_tfidf.replace(0, pd.NA).mean(axis=1) # Average TFIDF score of word
    if verbose: print(f"Runtime: {time.time() - start_time:.8f} s")
    else:
        pass # debugging traces below
        # clear_output(wait=True)
        # print(f"Last runtime: {time.time() - start_time:.8f} s")
    return df_tfidf

print('Prompt with high TFIDF:')
corpus = ["Give me five", "Five", "As a large language model I lack the ability to give five"]
print(corpus)
df_tfidf = tf_idf_summary(corpus, verbose=True)
display(df_tfidf)
print('Prompt with low TFIDF:')
corpus = ["Hello, I want you to assist me with a task", "Sure, go ahead", "I am here to assist you"]
print(corpus)
df_tfidf = tf_idf_summary(corpus, verbose=True)
display(df_tfidf)
print('Responses with low TFIDF (none):')
corpus = ["Tell me a naughty joke", "Why don't scientists trust atoms? Because they make up every"]
print(corpus)
```

[skippable] Metrics examples

```
tok_metrics_examples = df_tok_metrics.sample(n=1)
pd.set_option('display.float_format', lambda x: f'{x:.2f}')
display(tok_metrics_examples[['prompt', 'p_len', 'p_tc', 'p_chk_ratio']])
display(tok_metrics_examples[['response_a', 'ra_len', 'ra_tc', 'ra_chk_ratio']])
display(tok_metrics_examples[['response_b', 'rb_len', 'rb_tc', 'rb_chk_ratio']])
pd.reset_option('display.float_format')
```

	prompt	p_len	p_tc	p_chk_ratio	
46267	why do labia look so different	30.00	9.00	4.28	
	response_a	ra_len	ra_tc	ra_chk_ratio	
46267	There are a few reasons why labia can vary sig...	808.00	174.00	4.70	
	response_b	rb_len	rb_tc	rb_chk_ratio	
46267	The labia, which are part of the external fema...	1717.00	390.00	4.43	

Historical Roots: the CogPAST project (2016)

Early personal inspiration for conversational quality metrics (IT support agents)

- ❑ Ideated the '**Cognitive Satisfaction Score**' for as part of an AI (then called 'cognitive computing') hackathon back in 2016.
- ❑ Evaluated support desk responses via early **sentiment analysis** models (AlchemyAPI).
- ❑ Conceptual ancestor to SCBN-RQTL: measuring not what the agent says, but **how it fits the user needs**.

How to calculate a Cognitive Satisfaction Score

- The idea behind CogPAST is leveraging vast amounts of data as generated in a technical support ticketing system to measure the effectiveness of the support technician in transforming a negative attitude from the customer in the first message:

LANGUAGE: English

I received reporting from a customer that his user credentials were not accepted in the system and the "recover password" link is not having any effect, he still can't access the system by any means. The user account is xxxxxxxxxx@ibm.com. To my surprise, I've encountered the same issue after entering my credentials (xxxxxxx@ou.ibm.com) and also getting an error message (see attachment). When trying to reset my password I get the following message: You cannot change your password yet. Please wait at least 1 Day before changing your password again or contact IBM Customer Service (xxxxxxxsupport@ou.ibm.com) for assistance. This is not an issue of entering incorrect credentials, so I label it as P1 until we can use the system again and get a root cause explanation.

Click here to learn more about sentiment.

Entities	Sentiment
Keywords	negative

- Into a "positive" response or ser

LANGUAGE: English

Hi XXXXXXX, I tried the same after login in a dif was only noticed by me so we're fine by now.

Click here to learn more about sentiment.

Entities	Sentiment
Keywords	positive

How to calculate a Cognitive Satisfaction Score

- A simplistic determination of CogSS would use the sentiment score (SS) on the first customer message (left) and the score on the subsequent customer message (right):

This is just a very simplistic example to realize the potential of Alchemy API for our solution

Open Message

Score: -0.130343

Close Message

Score: 0.553684

- $\text{CogSS} = (\text{SS}(\text{close}) - \text{SS}(\text{open})) / (1 - \text{SS}(\text{open})) = (0.55 - -0.13) / (1 - -0.13) = +60\%$ "satisfaction improvement"
- When developing the score calculation, **other variables** will be taken into account, such as number of interactions until reaching resolution, sentiment profile of the customer based on other interactions and/or customer segment (customer is hard to satisfy, customer is "nice" ...), etc.